

Learning Discriminative Representations to Interpret Image Recognition Models

Thèse de Doctorat

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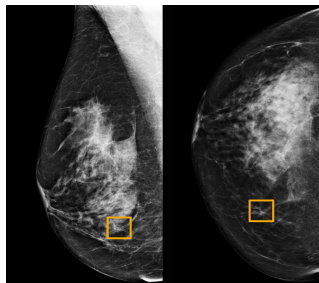
- Introduction
- 1 Opti-CAM: Optimizing saliency maps for interpretability
- 2 CA-Stream: Attention-based pooling for interpretable image recognition
- 3 A learning paradigm for interpretable gradients
- Closing Remarks

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- 1 Opti-CAM: Optimizing saliency maps for interpretability
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Introductory Examples

Consider these situations:



Motivation

Introductory Examples

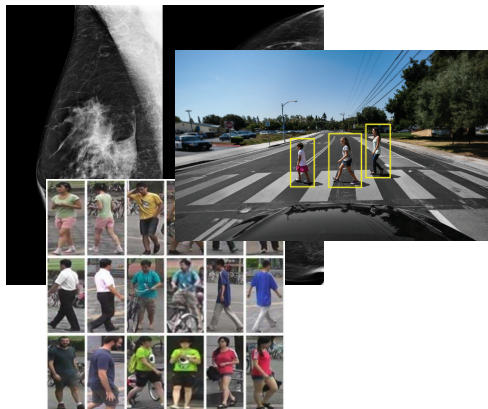
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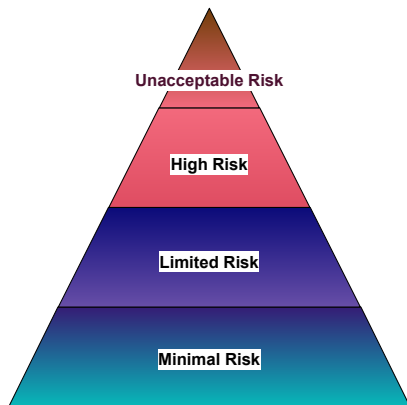
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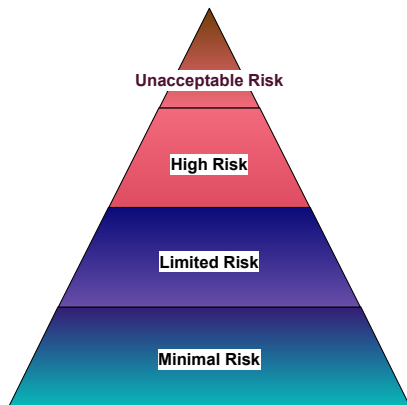
AI accepted in society?



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

Motivation

AI accepted in society?



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

We must **explain** the behaviour of these models and **interpret** their predictions.

Image Recognition Models

An ongoing revolution since 2012

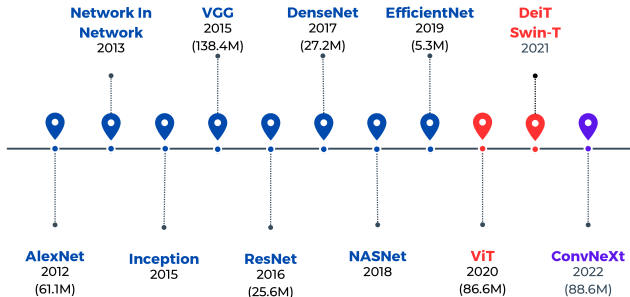
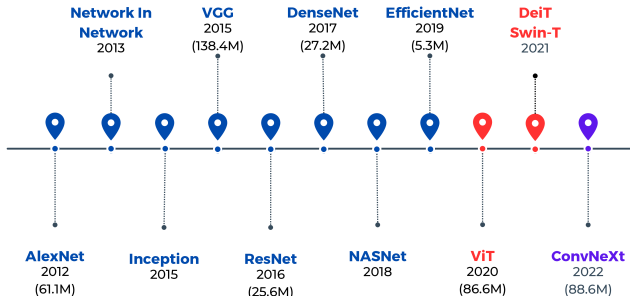


Image Recognition Models

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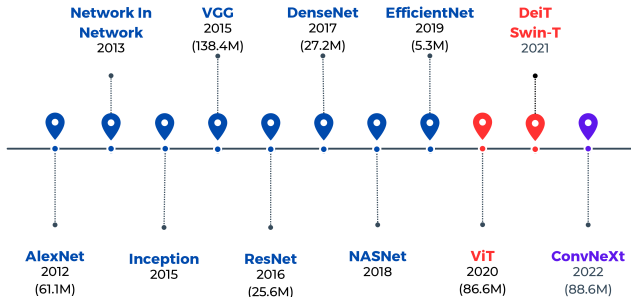


Contributing factors:

- Widespread of Internet.
- Big Dataset collection.
- Popularization of the GPU.

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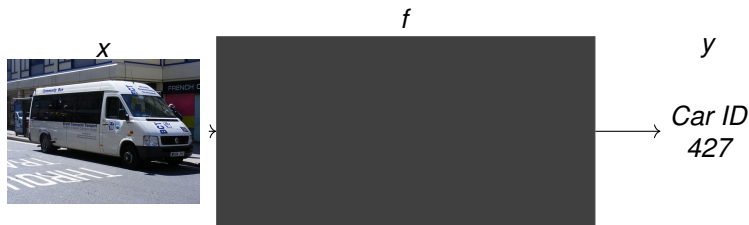
Two paradigms:

Convolutional Neural Networks (CNNs).

Transformers.

Motivation

We are interested in understanding models, behaving like a black box:

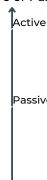


We want to *know why* $f(x) \rightarrow y$

A paradigm on interpretable studies

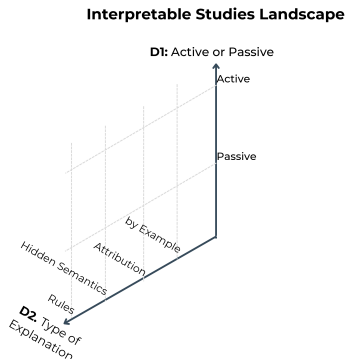
1. Passive or Active?

Interpretable Studies Landscape



A paradigm on interpretable studies

- ## 2. Type of Explanation
- Is it an example? An Attribution? Hidden Semantics? Or a Rule?*



Interpretability

A paradigm on interpretable studies

Interpretability approaches have been categorized by two studies [1, 2]:

1. **Passive or Active?**¹

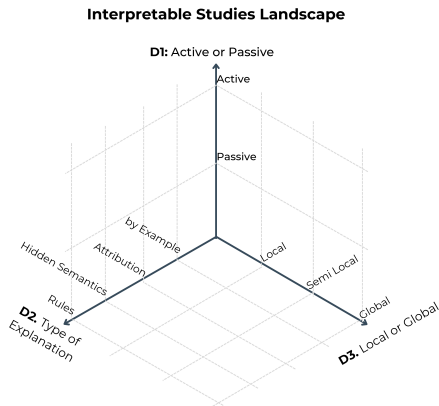
To add or not modifications to the model.

2. **Type of Explanation**

Is it an example? An Attribution? Hidden Semantics? Or a Rule?

3. **Local or Global?**

Is it a brief explanation? Does it cover more data/ of the model? Does it explain the model/data entirely?

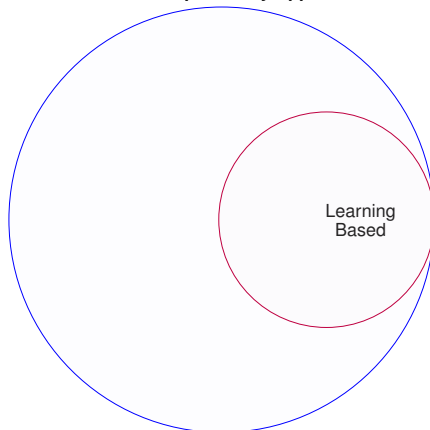


¹ Lipton defines Interpretability Similarly.
Transparency or Post-hoc Interpretations

Interpretability

Post-hoc Interpretability

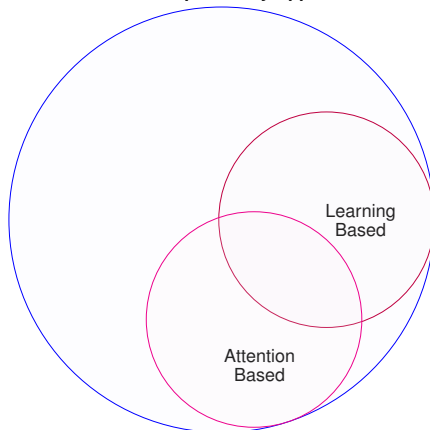
Post-hoc Interpretability Approaches



Interpretability

Post-hoc Interpretability

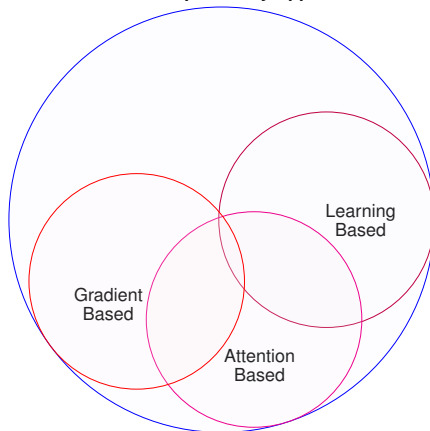
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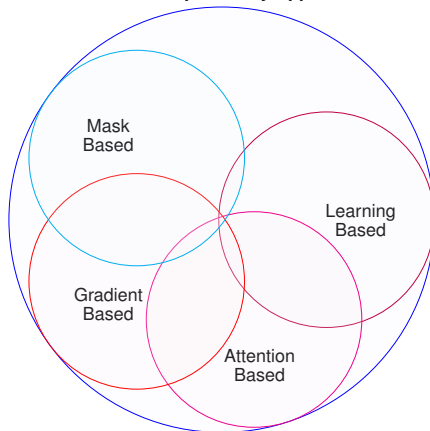
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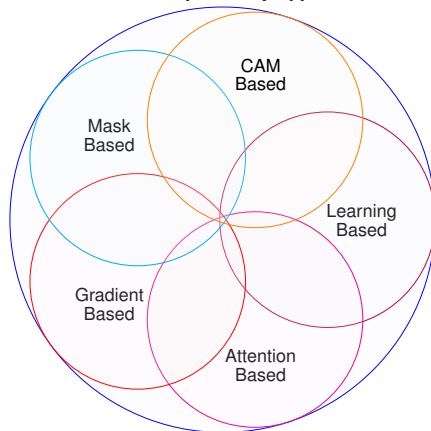
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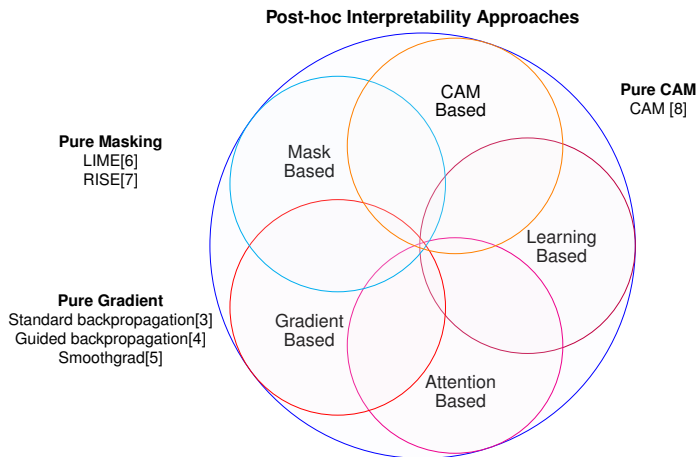
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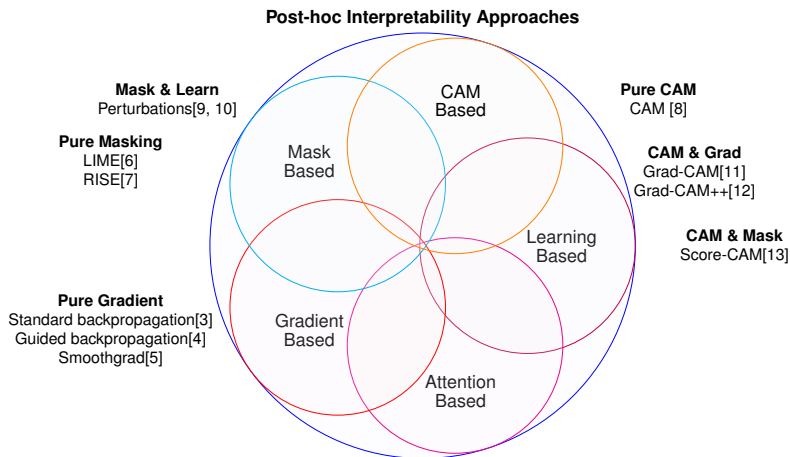


Post-hoc Interpretability



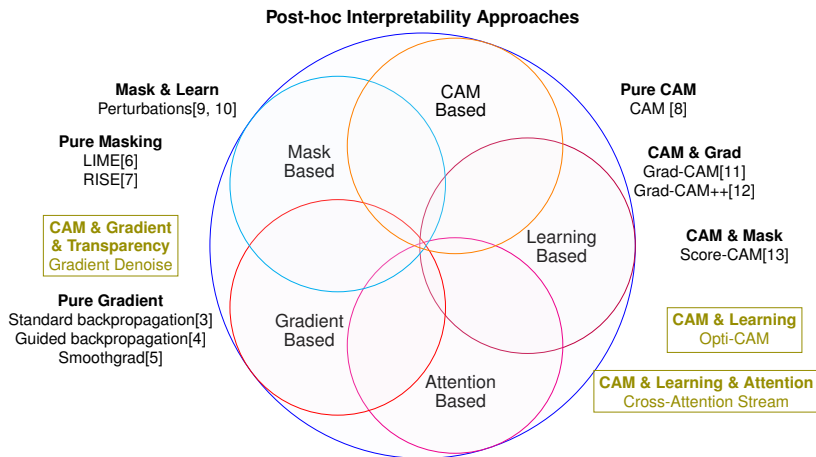
Interpretability

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Interpretability

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Contributions

Overview

We are interested in understanding models, behaving like a black box:



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Saliency Maps



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Gradients



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Other Representations



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Other Representations



We demonstrate improvements in these approaches

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Preliminaries

Premise

Is it possible to optimize a saliency map probability with the highest probability of prediction?



$$p(x)_{774} = 0.745$$



$$p(x)_{774}^{\text{CAM}} = 0.728$$

Preliminaries

Premise

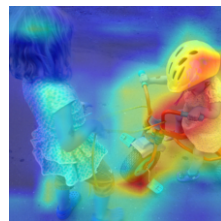
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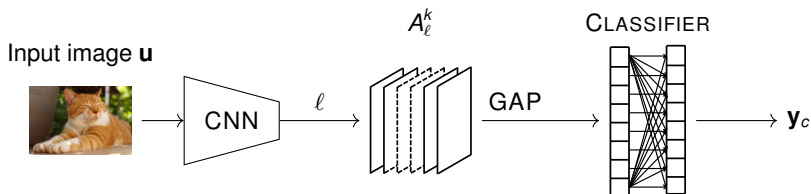
$$p(x)_{774}^{\text{CAM}} = 0.728$$



$$p(x)_{774}^{\text{Optimal}} = 0.853$$

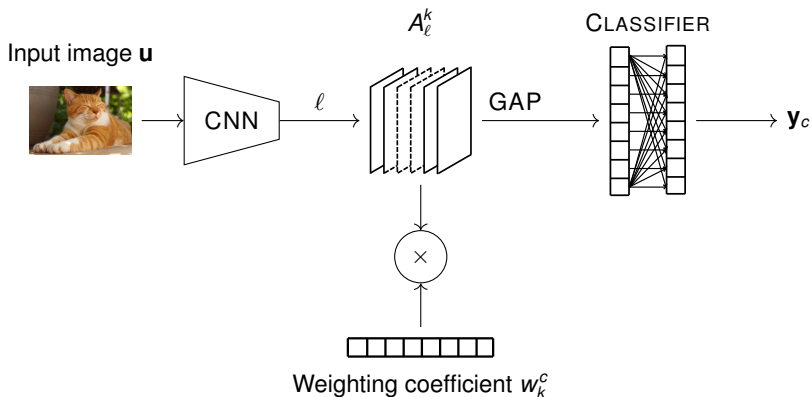
Background

CAM Methods



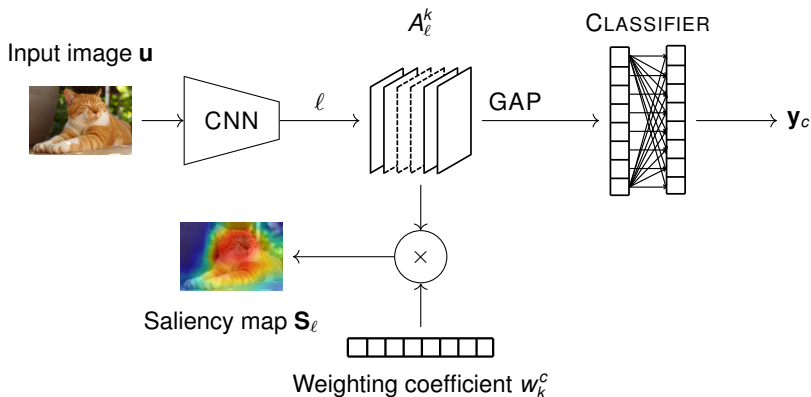
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CAM Methods



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CAM Methods



Background

CAM Formalized

For an activation function (h) and a feature map (A_ℓ^k):

$$S_\ell^c(\mathbf{u}) := h\left(\sum_k w_k^c A_\ell^k\right) \quad (1)$$

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Milestone CAM models:

CAM[8]:

w_k^c comes from the weights
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Grad-CAM/Grad-CAM++

[11, 12]:

w_k^c is computed using gradients flowing from prediction to input space. It can be used at any point of the model.

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w_k^c is computed from the contributions of individual feature maps masking the input image.

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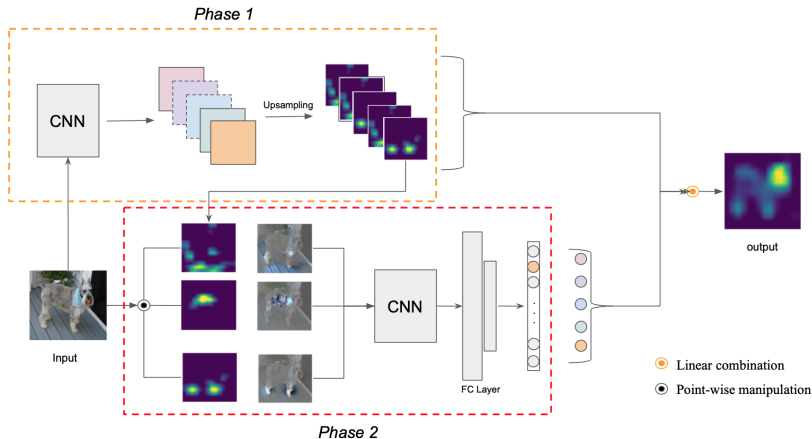
Score-CAM[13]:

w_k^c is computed from the contributions of individual feature maps masking the input image.

The approach found in Score-CAM is interesting. Diving deeper:

Background

Score-CAM



Original from [13]

Background

Score-CAM Formalized

Following equation (1):

Background

Score-CAM Formalized

Following equation (1):

- $h := \text{ReLU}$
- $\mathbf{w}_k^c := \text{softmax}(\mathbf{u}^c)_k$

Background

Score-CAM Formalized

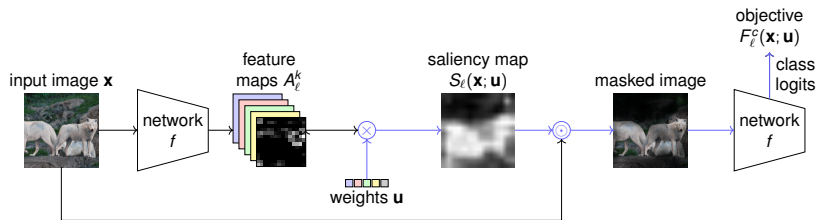
Following equation (1):

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With feature map contribution ($\mathbf{u}^c_k$), model (f), normalization (n) and upsampling (**up**):

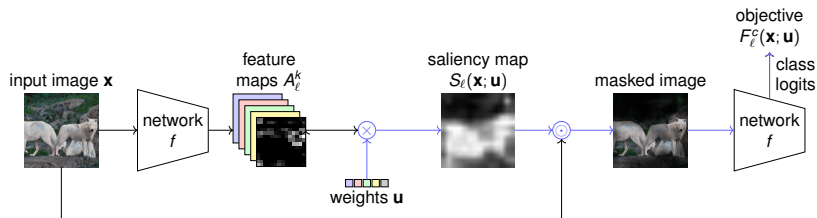
$$u_k^c := f(\mathbf{x} \odot n(\text{up}(A_\ell^k)))_c - f(\mathbf{x})_c \quad (2)$$

Opti-CAM Approach



Opti-CAM

Approach

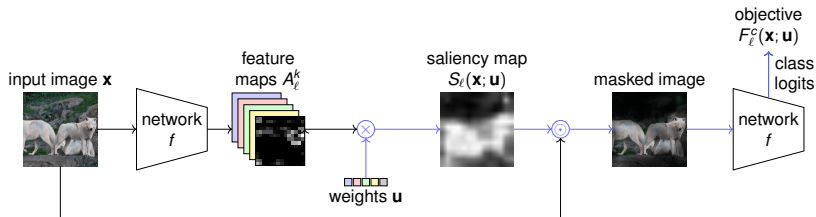


Consider S_ℓ , a function of $\mathbf{u}$ and $\mathbf{x}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

Opti-CAM

Approach



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$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

$\mathbf{u}^*$ is the vector that maximizes class c probability, when masking the the input with S_ℓ .

$$\mathbf{u}^* := \underset{\mathbf{u}}{\operatorname{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}) \quad (4)$$

And the objective $F_\ell^c(\mathbf{x}; \mathbf{u}) := f(\mathbf{x} \odot n(\text{up}(S_\ell(\mathbf{x}; \mathbf{u}))))$

Evaluating Interpretability

Three paradigms

Evaluating Interpretability

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- **Interpretable Image Recognition[12]**

Average Drop (AD), Average Increase (AI), Average Gain (AG).

Evaluating Interpretability

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- **Causality Analysis[7]**

Insertion & Deletion.

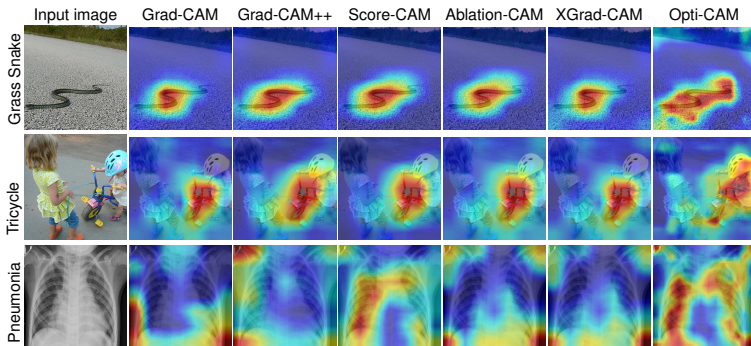
Evaluating Interpretability

Three paradigms

- **Interpretable Image Recognition[12]**
Average Drop (AD), Average Increase (AI), Average Gain (AG).
- **Causality Analysis[7]**
Insertion & Deletion.
- **Weakly Supervised Localization Approach**
Official Metric (OM), Localization Error (LE), Pixel-wise F_1 score (F1) Box Accuracy (BA), Standard Pointing Game (SP), Energy Pointing Game (EP).

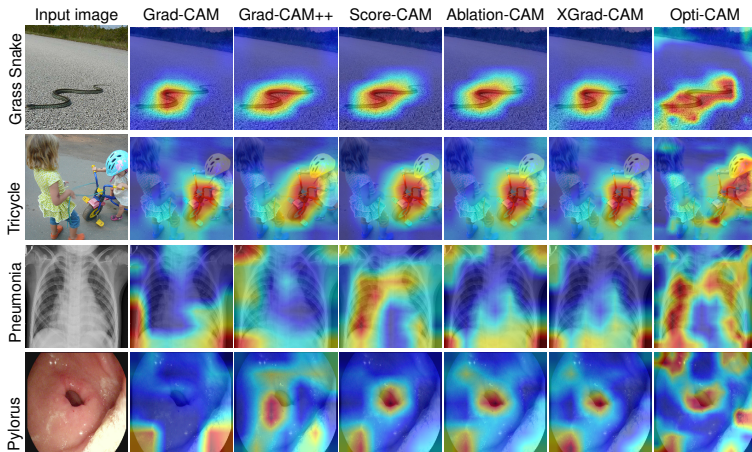
Results

Qualitative Evaluation



Results

Qualitative Evaluation



Results

Quantitative Evaluation

Classification Experiments:

METHOD	RESNET50						VGG16					
	AD↓	AG↑	AI↑	I↑	D↓	T(s)	AD↓	AG↑	AI↑	I↑	D↓	T(s)
Fake-CAM	0.8	1.6	46.0	50.7	28.1	0.00	0.5	0.6	42.6	46.1	26.9	0.00
Grad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	14.2	14.7	40.6	64.1	11.6	0.02
Grad-CAM++	12.9	16.0	42.1	66.0	14.7	0.03	17.1	10.2	33.4	62.9	12.2	0.02
Score-CAM	8.6	26.6	56.7	65.7	16.3	15.22	13.5	15.6	41.7	62.5	12.1	3.11
Ablation-CAM	12.5	16.4	42.8	65.9	14.6	18.26	15.5	12.6	36.9	63.8	11.4	2.98
XGrad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	13.8	14.8	41.2	64.1	11.7	0.02
Layer-CAM	15.6	15.0	38.8	67.0	14.2	0.08	48.9	3.1	13.5	58.3	6.4	0.07
ExPerturbation	38.1	9.5	22.5	70.7	15.0	152.96	43.0	7.1	20.5	61.1	15.0	83.20
Opti-CAM	1.5	68.8	92.8	62.0	19.7	4.15	1.3	71.2	92.7	59.2	11.0	3.94

Results

Quantitative Evaluation

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The optimization objective is well aligned with AD, AI, AG.
Insertion and Deletion are better suited for sparse saliency maps.

Results

Quantitative Evaluation

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Grad-CAM	72.9	65.8	49.8	56.2	69.8	33.3	1.30		71.1	62.3	42.0	54.2	64.8	32.0	1.39	
Grad-CAM++	73.1	66.1	50.4	56.2	69.9	33.1	1.29		70.8	61.9	44.3	55.2	66.2	32.3	1.38	
Score-CAM	72.2	64.9	49.6	54.5	68.7	32.4	1.25		71.2	62.5	45.3	58.5	68.2	33.4	1.40	
Ablation-CAM	72.8	65.7	50.2	56.1	69.9	33.1	1.26		71.3	62.6	43.2	56.2	65.7	32.7	1.39	
XGrad-CAM	72.9	65.8	49.8	56.2	69.8	33.3	1.30		70.8	62.0	41.9	53.5	64.4	31.6	1.41	
Layer-CAM	73.1	66.0	50.1	55.5	70.0	33.0	1.29		70.5	61.5	28.0	54.7	65.0	32.4	1.45	
ExPerturbation	73.6	66.6	37.5	44.2	64.8	38.2	1.59		74.1	66.4	37.8	43.3	62.7	36.1	1.74	
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Localization is hard. It acts more aligned to human interpretations than model ones.

Conclusions

- Classifier specific saliency maps. The probability of prediction of a class is the highest.
- No regularization needed unlike [9, 10]. Optimization only over feature dimensions.
- Localization is difficult for Opti-CAM. It is mostly human based.
- Average Gain complements the evaluation procedure.
- Published at Computer Vision and Image Understanding (CVIU) 2024.

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Preliminaries

Premise

Is it possible to incorporate transformers ideas onto CNNs, enhancing the predictive power of CAM explanations?



$$p = 0.7623$$



$$p = 0.7145$$

Preliminaries

Premise

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$$p = 0.7623$$



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$$p = 0.7478$$

Background

Self Attention

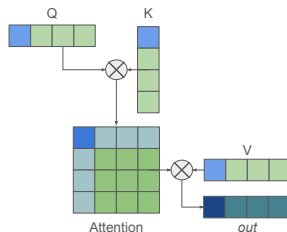
Self-attention is the building block of Transformers:

Background

Self Attention

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$$\text{Self Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (5)$$

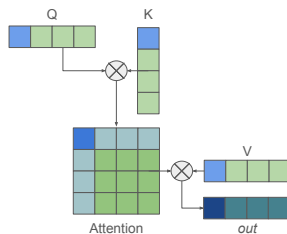


Background

Self Attention

Self-attention is the building block of Transformers:

$$\text{Self Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_e}} \right) V \quad (5)$$



An abstract class representation ([CLS]) updated with information from the embedding. [14]

Background

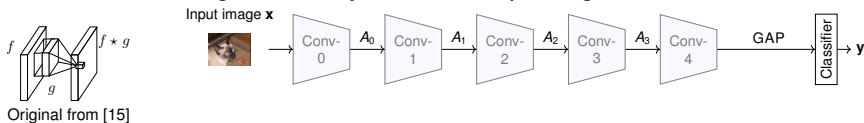
Convolutions and Self Attention

Different building blocks rely on different pooling mechanisms:

Background

Convolutions and Self Attention

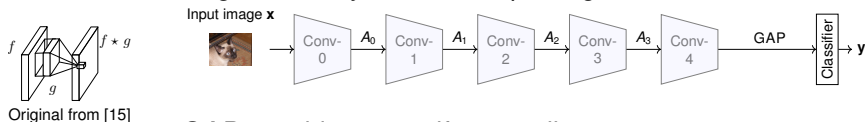
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Background

Convolutions and Self Attention

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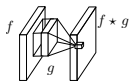


GAP considers an uniform pooling.

Background

Convolutions and Self Attention

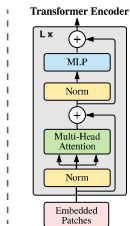
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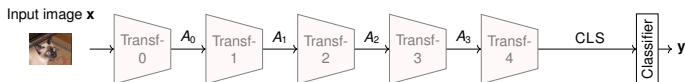
Original from [15]



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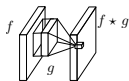
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Background

Convolutions and Self Attention

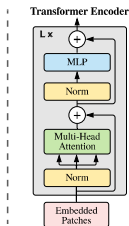
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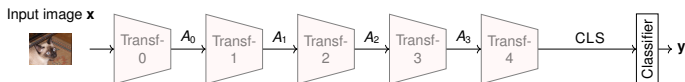
Original from [15]



GAP considers an uniform pooling.



Original from [16]



CLS is weighted with attention.

Motivation

Including CLS in CNNs

Cross Attention: Assuming an arbitrary embedding $\mathbf{q}$:

$$\text{CA}_\ell(\mathbf{q}_\ell, A_\ell) := A^\top \mathbf{q} = A_\ell^\top h_\ell(A_\ell \mathbf{q}_\ell), \in \mathbb{R}_\ell^d \quad (6)$$

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For a feature map A_ℓ , a pooled representation can be expressed via attention vector $\mathbf{a} = h_\ell(A_\ell \mathbf{q}_\ell)$:

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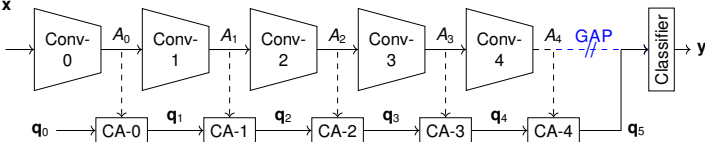
$$\text{CA}_\ell(\mathbf{q}_\ell, A_\ell) := A_\ell^\top \mathbf{a} \quad (7)$$

A pooling similar to GAP, masked in a fashion similar to CAM.

Cross Attention Stream

Given a model f we consider locations where critical operations take place:

Input image $\mathbf{x}$

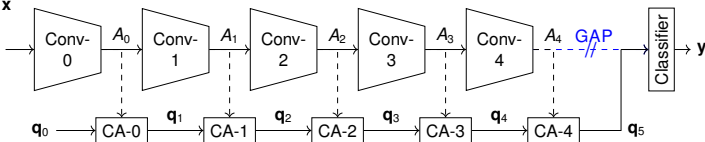


$$f = g \circ \text{GAP} \circ f_L \circ \dots \circ f_0 \quad (8)$$

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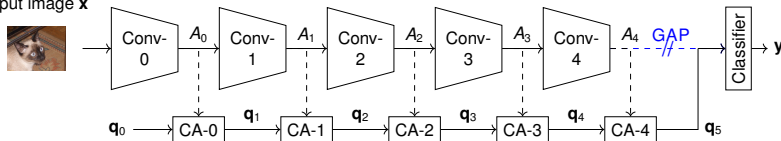
$$f = g \circ \text{GAP} \circ f_L \circ \dots \circ f_0 \quad (8)$$

A [CLS] token embedding is initialized as $\mathbf{q}_0 \in \mathbb{R}^{d_0}$.

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Given a model f we consider locations where critical operations take place:

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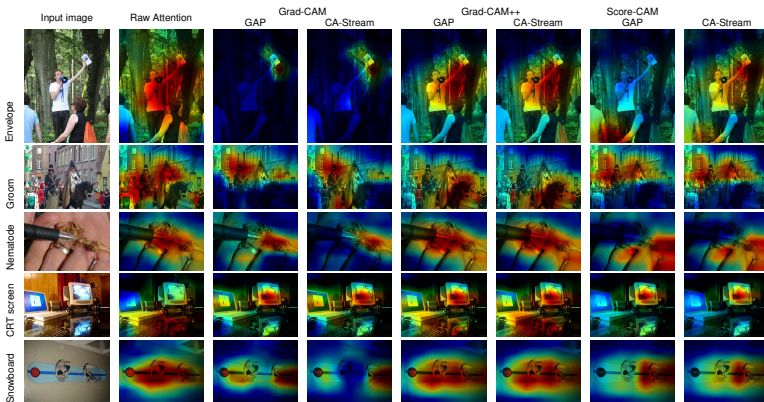
$$f = g \circ \text{GAP} \circ f_L \circ \dots \circ f_0 \quad (8)$$

A [CLS] token embedding is initialized as $\mathbf{q}_0 \in \mathbb{R}^{d_0}$. Updated with a sequence of embeddings $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$, interacting along a stream with features on stage ℓ :

$$\mathbf{q}_{\ell+1} = W_\ell \cdot \text{CA}_\ell(\mathbf{q}_\ell, A_\ell) \quad (9)$$

Results

Qualitative Experiments



Results

Quantitative Experiments

Recognition results

ACCURACY AND PARAMETERS					
NETWORK	POOL	GFLOPS	#PARAM	PARAM%	ACC↑
RESNET-18	GAP	3.648	11.69M	3.71	67.28
	CA	3.652	12.13M		67.54
RESNET-50	GAP	8.268	25.56M	27.27	74.55
	CA	8.288	32.53M		74.70
CONVNEXT-S	GAP	17.395	50.22M	1.95	83.26
	CA	17.400	51.20M		83.14
CONVNEXT-B	GAP	30.747	88.59M	1.96	83.72
	CA	30.753	90.33M		83.51

CA Maintains recognition properties.

Results

Quantitative Experiments

NETWORK	METHOD	POOL	AD↓	AG↑	AI↑	I↑	D↓
RESNET-50	Grad-CAM	GAP	13.04	17.56	44.47	72.57	13.24
		CA	12.54	22.67	48.56	75.53	13.50
	Grad-CAM++	GAP	13.79	15.87	42.08	72.32	13.33
		CA	13.99	19.29	44.60	75.21	13.78
	Score-CAM	GAP	8.83	17.97	48.46	71.99	14.31
		CA	7.09	23.65	54.20	74.91	14.68
CONVNEXT-B	Grad-CAM	GAP	33.72	2.43	15.25	52.85	29.57
		CA	19.45	13.96	32.89	86.38	45.29
	Grad-CAM++	GAP	34.01	2.37	15.60	52.83	29.17
		CA	36.69	8.00	21.95	85.39	53.42
	Score-CAM	GAP	43.55	2.23	15.67	50.96	39.49
		CA	23.51	11.04	27.35	83.41	60.53

Results

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NETWORK	METHOD	POOL	AD↓	AG↑	AI↑	I↑	D↓
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	Grad-CAM++	GAP	34.01	2.37	15.60	52.83	29.17
		CA	36.69	8.00	21.95	85.39	53.42
	Score-CAM	GAP	43.55	2.23	15.67	50.96	39.49
		CA	23.51	11.04	27.35	83.41	60.53

CA obtains stronger representations for interpretability.
Deletion poses issues due OOD data.

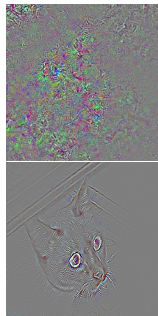
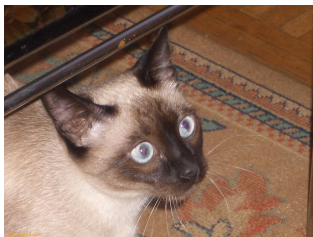
Conclusions

- CAM behaves like Attention. Attention on final layers ought to behave like CAM.
- Pooling matters. Smarter pooling yields improved interpretations, maintaining recognition performance.
- Deletion requires a revisit. Zeros on images still provide a signal.
- Published in the XAI Workshop at CVPR 2024.

Preliminaries

Premise

Can smoothing gradients during training, improve interpretability properties?



Background

Leveraging Backpropagation

The response (y) of a model (f), after forwarding an input x can be visualized on the input space, via backpropagation:

$$\frac{\partial x}{\partial y} = \frac{\partial x}{\partial F_0} \frac{\partial F_0}{\partial F_1} \cdot \dots \cdot \frac{\partial F_{-1}}{\partial y} \quad (10)$$

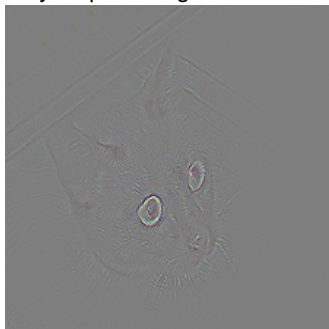
Background

Soft Gradients

Soft gradients can be achieved by two approaches:

Guided Backpropagation [4]

Achieved by constraining backprop to only let positive gradients flow:



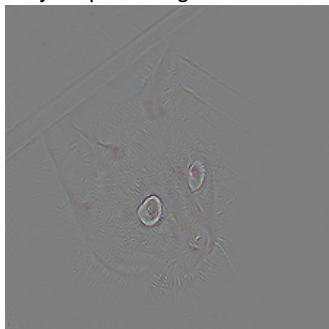
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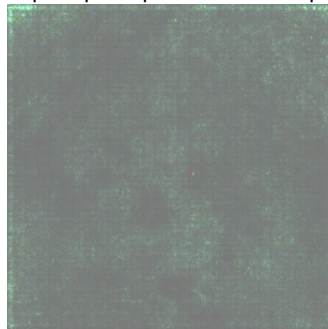
Guided Backpropagation [4]

Achieved by constraining backprop to only let positive gradients flow:



Smoothgrad [5]

Achieved by iteratively adding noise to the input space prior to forward pass:



Interpretable Gradients

Formalized

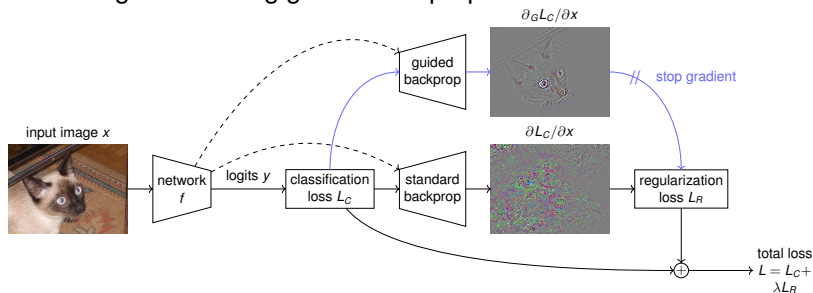
Goal: Denoise the gradient during training using guided backprop.
How? Regularize using guided backprop.

Interpretable Gradients

Formalized

Goal: Denoise the gradient during training using guided backprop.

How? Regularize using guided backprop.



Interpretable Gradients

Smooth Gradient as Regularization

Assuming target labels T :

$$L_R(X, \theta, T) = \frac{1}{n} \sum_{i=1}^n E(\delta x_i, \delta_G x_i), \quad (11)$$

Interpretable Gradients

Smooth Gradient as Regularization

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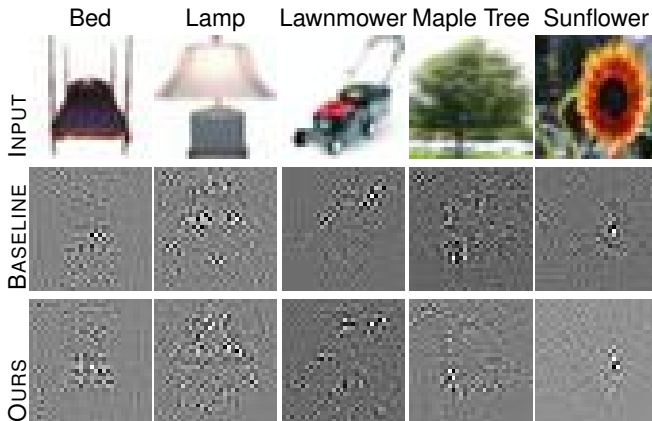
And the total loss:

$$L(x, \theta, t) = L_C(x, \theta, t) + \lambda L_R(x, \theta, t), \quad (12)$$

Results

Qualitative Experiments

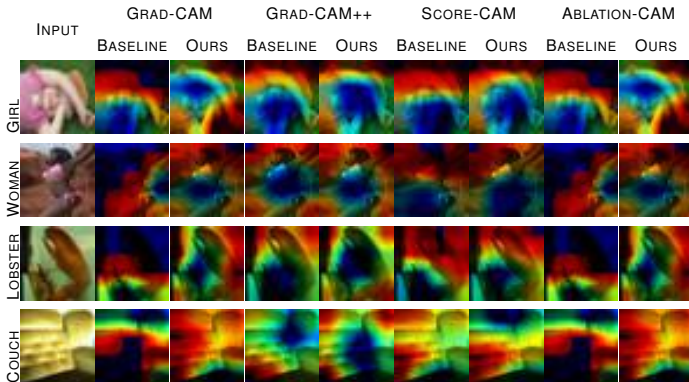
Denoising effect



Results

Qualitative Experiments

CAM Visualizations



Results

Quantitative Experiments

Recognition Metrics

RECOGNITION METRICS			
MODEL	ERROR	λ	Acc $\uparrow$
RESNET-18	-	-	73.42
	COSINE	7.5×10^{-3}	72.86
MOBILENET-V2	-	-	59.43
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Recognition properties are maintained.

Results

Quantitative Experiments

Interpretability metrics

INTERPRETABLE RECOGNITION METRICS						
RESNET-18						
METHOD	L_R	AD↓	AG↑	AI↑	INS↑	DEL↓
GRAD-CAM	-	30.16	15.23	29.99	58.47	17.47
	COSINE	28.09	16.19	31.53	58.76	17.57
GRAD-CAM++	-	31.40	14.17	28.47	58.61	17.05
	COSINE	29.78	15.07	29.60	58.90	17.22
SCORE-CAM	-	26.49	18.62	33.84	58.42	18.31
	COSINE	24.82	19.49	35.51	59.11	18.34
ABLATION-CAM	-	31.96	14.02	28.33	58.36	17.14
	COSINE	29.90	15.03	29.61	58.70	17.37
AXIOM-CAM	-	30.16	15.23	29.98	58.47	17.47
	COSINE	28.09	16.20	31.53	58.76	17.57

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AXIOM-CAM	-	30.16	15.23	29.98	58.47	17.47
	COSINE	28.09	16.20	31.53	58.76	17.57

Interpretable properties are enhanced. Deletion still poses an issue.

Conclusions

- Denoising CNN gradients improve interpretability properties, maintaining recognition.
- Negative gradients contribute to stability during training.
- Resolution matters: small inputs require shallow layer targeting.
- Published in VISAPP 2024.

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- 1 Opti-CAM: Optimizing saliency maps for interpretability
- 2 CA-Stream: Attention-based pooling for interpretable image recognition
- 3 A learning paradigm for interpretable gradients
- Closing Remarks

Closing Remarks

Conclusion

- Context is important for a prediction. Salient information is found in the background.
- Pooling matters. Small changes lead to enhanced interpretations, with little penalty to predictive power.
- Noisy gradients are a byproduct of training. Negative values act as regularizers.
- Insertion & Deletion are problematic. OOD data affects the classifier.

Closing Remarks

Perspectives

- Less complex approaches are desired. Optimizing gradient denoise is an desired.
- A thorough reality check for *fair* comparison is desired across methods. Set the stage for future comparisons.
- Explanations should extend beyond *groundtruth*. The reality check could include such scenarios.
- Deep models are growing in complexity. Designing approaches to interpret transformers is one option.
- Interpretability must be extended to high impact domains: Medicine, Surveillance, Autonomous Vehicles.
- *For me?*

Closing Remarks

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- Interpretability must be extended to high impact domains: Medicine, Surveillance, Autonomous Vehicles.
- *For me?* I want to pursue a career in research, either in academy or industry.

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References I



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